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RESEARCH ARTICLE

Mole-inspired Forepaw Design and Optimization Based on Resistive Force Theory

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Abstract
Moles exhibit highly effective capabilities due to their unique body structures and digging techniques, making them ideal models for biomimetic research. However, a major challenge for mole-inspired robots lies in overcoming resistance in granular media when burrowing with forelimbs. In the absence of effective forepaw design strategies, most robotic designs rely on increased power to enhance performance. To address this issue, this paper employs Resistive Force Theory to optimize mole-inspired forepaws, aiming to enhance burrowing efficiency. By analyzing the relationship between geometric parameters and burrowing forces, we propose several forepaw design variations. Through granular resistance assessments, an effective forepaw configuration is identified and further refined using parameters such as longitudinal and transverse curvature. Subsequently, the Particle Swarm Optimization algorithm is applied to determine the optimal forepaw design. In force-loading tests, the optimized forepaw demonstrated a 79.44% reduction in granular lift force and a 22.55% increase in propulsive force compared with the control group. In robotic burrowing experiments, the optimized forepaw achieved the longest burrow displacement (179.528 mm) and the lowest burrowing lift force (0.9355 mm/s), verifying its effectiveness in reducing the lift force and enhancing the propulsive force.

Keywords Resistive force theory · Mole-inspired forepaw design · Structural optimization · Bioinspired robot

1 Introduction

Burrowing robots hold potential for advancing mechanized agricultural production and addressing demands in space exploration missions [1]. However, enabling effective burrowing in Granular Medium (GM) presents a significant technical challenge. Unlike wheeled robots, where only specific moving parts contact GM, burrowing robots must be fully buried within the medium. This requires a traction force generated through the interaction between the burrowing module and GM to facilitate internal movement. Such interaction involves a complex robot-soil dynamics [2, 3]. Optimizing the geometry of the burrowing module can reduce the mechanical complexity encountered by the robot during the burrowing process, thereby improving its efficiency and maneuverability.

The design of burrowing robots is largely inspired by natural burrowing organisms [4], with earthworms, plant-roots, and moles being prominent models in recent research [5]. These studies aim to translate the burrowing abilities of these organisms into functional robot designs, thereby enhancing robotic performance to a level comparable with their biological counterparts. In particular, mole-inspired burrowing robots emulate the burrowing techniques of moles to improve robot performance with in GM, providing insights into optimized burrowing mechanics.

Recent research on mole-inspired burrowing robots has yielded promising developments. Takashi et al. [6] designed a cam mechanism for burrowing module overturning, enabling a robot to burrow 150 mm in wet soil and effectively replicate the forelimb burrowing action of moles. However, this design still faces limitations in horizontal

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movement within GM remains limited. Christian et al. [7] developed a mole-inspired robot with a hybrid auger-forelimb mechanism, where the auger crushes soil and the forelimbs clear it to facilitate movement. However, its burrowing performance has not been substantiated by experimental data. Lee et al. [8] proposed a hybrid mechanism for a mole-like robot featuring a rotating serrated cutter for breaking hard rocks and a linkage mechanism in the forelimbs to clear crushed soil. This robot moves on tracks, which increases its mass and energy consumption. Despite these innovative approaches, most mole-inspired robots still struggle to achieve the burrowing efficiency observed in real moles. Many studies do not address the resistance forces encountered in GM or provide comprehensive designs for the burrowing module. Some mole-inspired robots even omit specialized burrowing modules, instead relying on drill bits and increased penetration force to enhance performance. These limitations underscore the ongoing challenge of designing effective burrowing modules, which necessitates a deeper understanding of robot-soil interaction.

Recent advances in robot-soil interaction have extended fluid resistance theory to GM, establishing a relationship between the geometric parameters of intruders and the resistance force. This approach, known as granular Resistive Force Theory (RFT), has successfully characterized force interactions within GM, yielding promising experimental results [9–11]. By conducting experiments to obtain relevant RFT parameters for GM, a mathematical model of burrowing motion can be developed, which can guide the design of burrowing modules in mole-inspired robots. Furthermore, applying optimization algorithms to refine the module parameters may significantly enhance robotic burrowing performance [12].

This paper applies RFT to establish the force-loading relationship for the burrowing module in mole-inspired robots. RFT is first applied to quantify the burrowing performance of the mole forelimb. Based on biological analysis of the mole forepaw, a parametric model for the robotic forepaw is developed. An optimization objective function is developed using mechanical indices, and Particle Swarm Optimization (PSO) is employed to optimize forepaw parameters. Compared with a conventional rectangular-plate forepaw, the optimized design enhances thrust force while reducing lift force, resulting in superior burrowing performance. These improvements are validated experimentally.

The outline is organized as follows: Sect. 2 briefly describes the RFT, experimental setup and validates the RFT. In Sect. 3, we analyze the structural design of a mole forelimb forepaw and use the RFT as a design guide to optimize the mole-inspired forelimb forepaw with the aim of improving the mechanical property. The optimized mole-inspired forepaw is then validated in subsection D of Sect. 3 using a robotic forelimb burrowing in GM (polypropylene fine granular).

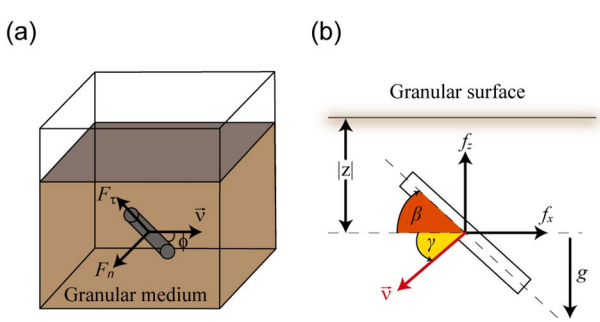

Fig. 1 Concept of RFT modeling. **a** RFT in the horizontal plane; **b** RFT in the vertical plane

2 RFT Analysis Method and Model Validation

2.1 Granular Resistive Force Theory

In contrast to fluid dynamics, where forces are calculable via the Navier–Stokes equations [13], a comprehensive mechanical model for an intruder in GM is remains underdeveloped. Recent research has utilized the Discrete Element Method (DEM) [14] to model forces between GM and burrowing modules. Although DEM provides precise force predictions, it entails high computational costs and limitations the granular bed size.

Li et al. [10] drew inspiration from the drag model developed for analyzing small lizards swimming horizontally in GM [9] and proposed the planar granular RFT. This theory divides an intruder with complex morphology and kinematics into discrete sub-elements, each assumed to generate an independent force. The cumulative force on the intruder is calculated by summing the forces on each subpart in GM [15]. However, RFT is applicable only for quasi-static movements where intruder velocity remains below 50 cm/s; at these velocities, the granular force is friction-dominated and inertia forces are negligible [16, 17]. RFT is defined as follows:

$$F_{total} = \int_{surf} \alpha_{x,z}(\beta, \gamma)\, |z|\, \mathrm{ds} \tag{1}$$

where F_{total} represents the total force exerted on the intruder by GM, *ds* is the surface area of an individual subpart of the intruder, and *z* represents depth. The inclination angle β and attack angle γ define the orientation and movement direction of the intruder within GM (Fig. 1). The function $\alpha_{(x,z)}$ (β, γ) depends on β and γ and varies according to the type of GM and the surface interaction with the intruder,

necessitating experimental calibration. This force prediction method demonstrates improved accuracy in dry GM.

Notably, when the direction of the intruder velocity is parallel to the horizontal plane, RFT requires adaptation into an empirical form for modeling [11]:

$$F = \int (d\mathbf{F_t} + d\mathbf{F_n}) = \int 2k\rho g r\,|z|\,\{f_n(\mathrm{v}\cdot\mathrm{n})\mathrm{n} + (\mathrm{v}\cdot\mathrm{t})\mathrm{t}\}\,\mathrm{ds} \tag{2}$$

where $k\rho g z$ is the effective hydrostatic pressure at depth z, with ρ defined as the particle density multiplied by the volume fraction. For loosely stacked granular material with particle diameter 0.3 mm, $k = 2.5$. Here, r is the cylinder radius, and the matrix plate is half of the plate thickness. The normal direction n and tangential direction t are also specified, where $\mathbf{F_n}$ is associated with the angle ϕ [9, 11]. The anisotropy factor for normal force takes the following form for the GM in consideration, with parameters $C = 1.8$ and $\varphi_0 = 13.8°$ [11]:

$$f_n(\mathrm{v}\cdot\mathrm{n}) = 1 + \frac{C}{\sqrt{\tan^2\varphi_0 + (\mathrm{v}\cdot\mathrm{n})^2}} \tag{3}$$

2.2 RFT Model Validation

RFT is particularly applicable to dry GM. In this study, polypropylene fine granules serve as substitutes for planetary soil simulants. Comparative tests are conducted under identical compaction conditions by ensuring the granules dry (Table 1). To establish the RFT parameters for polypropylene granules, a test bench is constructed for calibration (Fig. 2a). The setup includes a 25 mm × 30 mm × 2 mm 3D-printed experiment plate, a 6-axis robotic arm (UR3), a 3-axis force sensor (LA3D40), a PC data acquisition system, and a plastic container (300 mm × 400 mm × 600 mm) filled with polypropylene granules, maintaining a fixed granular bed height.

In one experiment, the robotic arm moves the experimental plate through the GM at a controlled speed of 10 mm/s in different directions. Each experiment is replicated three times to mitigate random errors. Before each trial, the granule bed is leveled using a brush, and the plate is positioned to start and end away from the container edges to minimize edge effects. Data are collected for $\gamma \in \{0°, 30°, 60°\}$, $\beta \in \{30°, 45°, 60°, 75°, 90°\}$ and $z \in$ (30 mm, 60 mm). The calibration parameter functions for the GM are calculated by Eq. 1, with results shown in Fig. 2c, d.

The calibrated parameters are substituted into the RFT to compute F_{z_sim} and F_{x_sim}. A set of experimentally measured F_{z_data} and F_{x_data} values is selected for comparison (with $\beta = 90°$ and $\gamma = 60°$), and the maximum and average errors are calculated. As shown in Fig. 2b, the maximum errors of F_x and F_z are 9.57% and 9.45%, respectively, while the average errors are 3.15% and 4.66%. These results indicate that the RFT model can effectively predict the resistance forces in GM.

Table 1 Parameters of the polypropylene fine granule

Parameter	Value	Unit
Granular sizes	0.6–0.8	mm
Bulk densities	0.45	g/cm^3
Granular densities	0.95	g/cm^3
Resting angle	32–34	°
Poisson ratio	0.3	–
Volume fraction	58–62%	–

3 Design and Optimization of Claw Toe Configurations

3.1 Design and Optimization of Forepaw Configurations

Biomimetic studies indicate that organisms have developed advanced functionalities through evolutionary adaptation to their environments over time [21, 22]. Burrowing animals, in particular, have evolved limbs with specialized mechanical properties that reduce cutting resistance during burrowing, providing valuable models for the biomimetic design

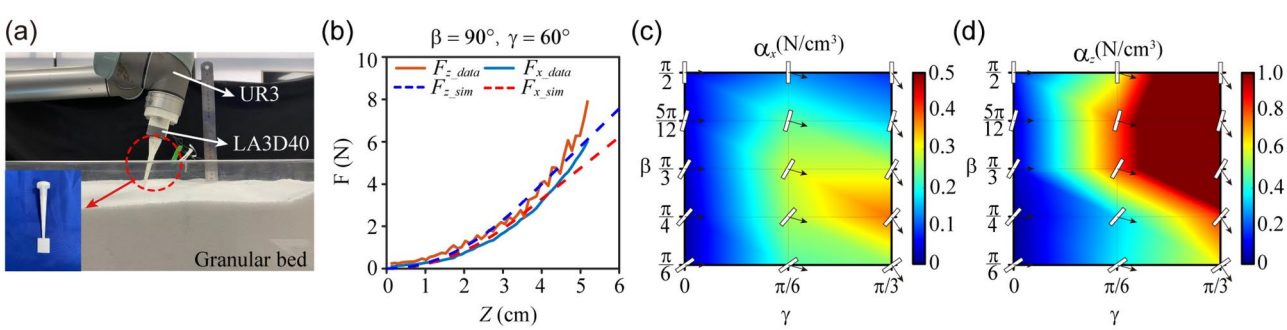

Fig. 2 Calibration experiments for RFT parameters using polypropylene fine granules. **a** Experimental platform; **b** validation experiment for Calibrated multipliers; **c** variation range of the calibrated multiplier α_x; **d** variation range of the calibrated multiplier α_z

of soil-cutting tools [21, 23, 24]. The mole (*Scaptochirus moschatus*) [25], a typical example of a highly efficient burrower, demonstrates high burrowing performance due to the unique structural adaptations of its forelimbs (Fig. 3a). Specifically, its forepaw toes are well-suited for soil cutting, making it an ideal model for designing soil-cutting and burrowing modules. Additionally, the bowl-shaped forepaw helps collect and clear soil debris, offering valuable insights for designing high-performance burrowing modules.

The forepaw of a forelimb-burrowing mole consists of metacarpal bones and six phalanges, with the sixth phalanx being underdeveloped and forming a tiny bone [19]. The functional forepaw primarily comprises the metacarpals and five fully developed phalanges (Fig. 3b) [26]. Its overall elliptical shape is, along with skin folds that assist in soil collection, allows the forepaw to form an bowl-like structure. Studies by Tong et al. [25] reveal that the mole toe geometry, particularly the smooth, nearly invariant longitudinal curvature, helps maintain a flat inner channel wall during burrowing (Fig. 3c). The longitudinal and transverse contour curves of the toes exhibit a convex structure, with the transverse curvature being more pronounced at the middle and end, increasing contact stress with the soil to aid penetration. The widening of the middle and end of toe also supports tunnel expansion during burrowing [27, 28]. Forelimb-burrowing moles primarily use the second and third toes (Fig. 3b) to cut through soil and remove debris along an elliptic burrowing path [29]. From these, the design requirements for a robotic burrowing forepaw can be summarized as follows: (i) the palm and toes should have smooth and continuous surfaces; (ii) the second toe should serve as a reference for configuring all other toes; (iii) the metacarpal region should be broad, bowl-shaped, and exhibit a specific curvature.

In addition to the design principles, dimensional parameters must be carefully considered. The structural parameters of the first-generation mole-inspired burrowing robot (Fig. 4a) serves as a reference, with a detailed description of the robot provided in chapter III, Sect. 2. Table 2 outlines the dimensional constraints, ensuring that different forepaw shapes maintain consistent thickness, length, width, and curvature to determine the optimal design. For added flexibility, the number of toes on the robotic forepaw is set to 3.

3.2 Forces Calculation and Comparison for Different Forepaws

Firstly, five different forepaw designs were created according to specified size constraints and design criteria (Fig. 4b). Each of these designs were paired with one of three bionic toe structures modeled after the contour curves of the second toe. To simplify the optimization process, the established RFT was used to predict the granular forces acting on each forepaw. Based on predefined performance metrics, an initial screening identified the designs most suitable for further detailed parameter optimization. It is important to note that forepaw curvature optimization was reserved for subsequent stages, thus the initial designs were uniformly planar, with all dimensions limited to 45 mm in length and 35 mm in width.

Unlike quadrupedal robots, which aim to maximize the vertical force F_z and thrust F_x for surface movement in GM [30], burrowing robots must fully engage with the media and navigate complex, non-linear robot-soil contact forces. For a burrowing robot to achieve locomotion within

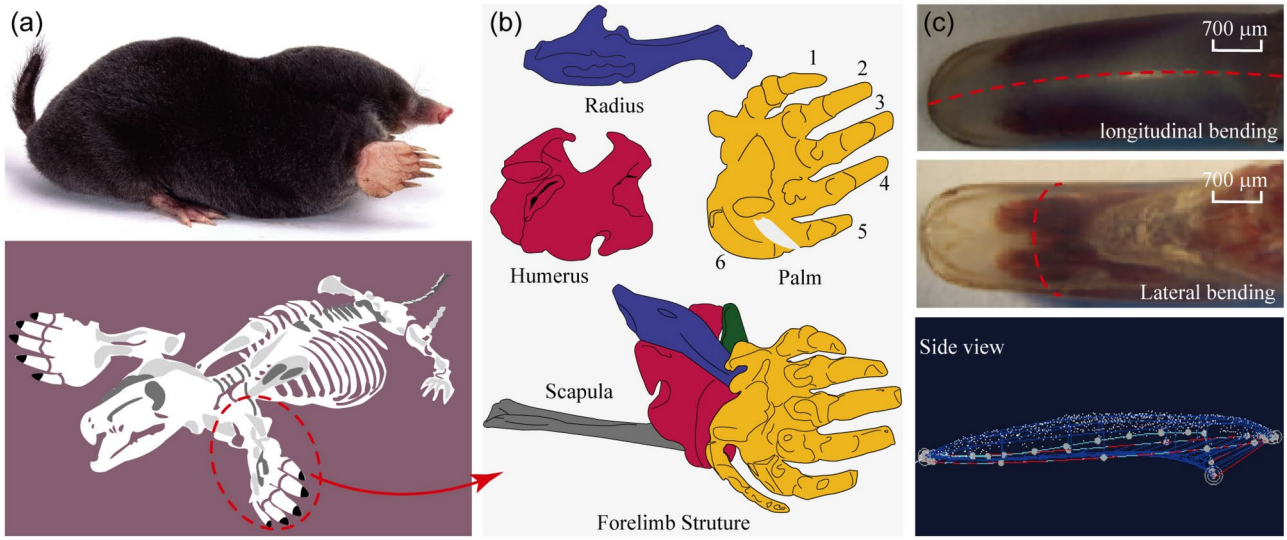

Fig. 3 Forelimb-burrowing mole. **a** Internal skeletal structure and biomorphology [18]; **b** skeletal structure of the forelimb and number of phalanges [19]; **c** frontal view of the second phalanx toe, reverse view, and contour curve of toe shape [20]

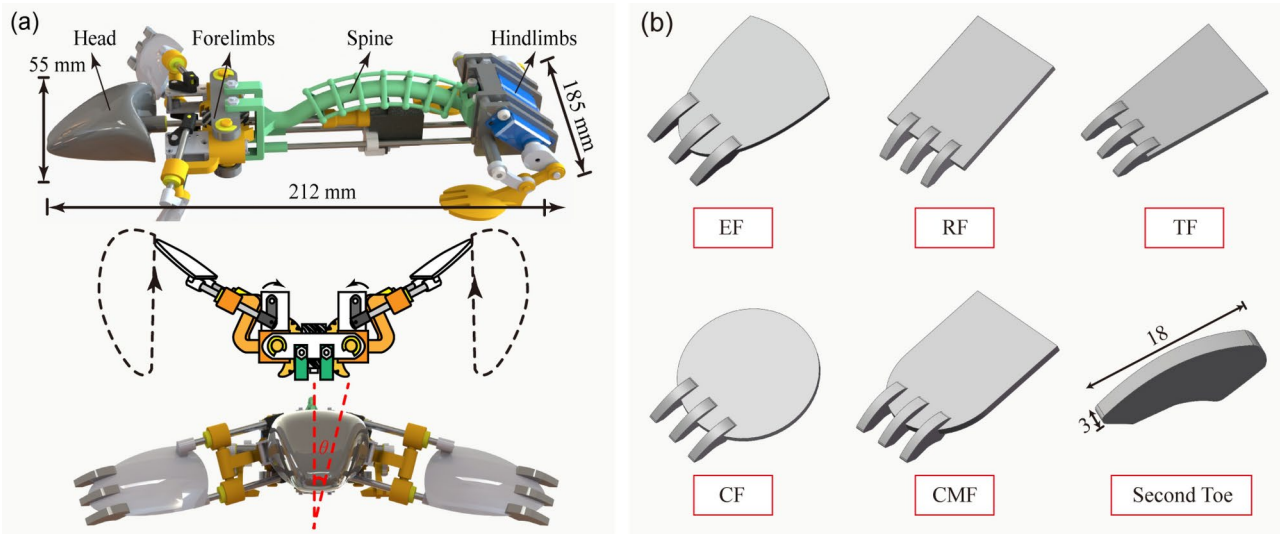

Fig. 4 Mole-inspired forelimb-burrowing robot for experiments and its five forepaw shapes. **a** Burrowing trajectory and definition of θ in the front view of the robot; **b** Forepaw designs: elliptical Forepaw (EF), Rectangular Forepaw (RF), Trapezoidal Forepaw (TF), Circular Forepaw (CF), and Circular Moment Mixed Forepaw (CMF), alongside the second toe bone dimensions (length: 18 mm, width: 3 mm)

Table 2 Parameters of the mole-inspired burrowing robot forepaw

Parameter	Value	Unit
Length	40–50	mm
Width	30–40	mm
Thickness	2	mm
Longitudinal curvature	60–80	°
Transverse curvature	20–40	°

a granular bed, it must first overcome the net upward force exerted by the granules while simultaneously generating sufficient thrust to propel itself forward. Therefore, the primary screening criterion for a forepaw structure is its ability to minimize F_z while maximizing F_x during movement through granules. In this simulation, the velocity for each forepaw is set at 10 mm/s, with the depth from the forepaw center of mass to the surface maintained at 4 cm.

When the robot engages in horizontal burrowing, the granular resistance-represented by the normal force F_n and tangential force F_t-acting on the five different forepaw designs can be calculated using Eq. 2. Given that the mole-inspired robot (Fig. 4a) employs a reciprocating burrowing motion, the angle ϕ between the forepaw and the direction of velocity varies periodically. During the cutting phase, ϕ gradually decreases. Therefore, the forces on each forepaw are analyzed at ϕ values of 0°, 30°, and 60° (Fig. 5a–c). The results indicate that as ϕ increases, F_t for each forepaw initially rises and then declines. This pattern occurs because, at the onset of movement, the forepaw encounters greater granular resistance while breaking the tight connections between granules. Once the granules are loosened, the forepaw transitions into a debris removal phase, resulting in reduced resistance. Notably, both the EF and CMF experience less granular resistance than the other three forepaw designs across different ϕ values, with the EF exhibiting the least resistance overall.

In contrast, when there is a non-zero angle between the forepaw surface and the granular bed in the direction of gravity, Eq. 1 must be used to calculate the granular resistance, expressed as F_z (lift force) and F_x (horizontal force), for each forepaw. Theoretically, increased horizontal resistance from the granules can yield greater traction force for the robot during quasi-static motion. However, excessive lift force from the granules can impede the movement of the robot within the medium. As shown in Fig. 2c, d, the value of $\alpha_z(\beta, \gamma)$ is lower when $\beta = \{30°, 45°, 60°\}$ and $\gamma = 0°$, indicating that the forepaw experiences reduced F_z under these conditions. To compare the granular resistance of the five forepaw designs, tests were conducted at $\beta = \{30°, 45°, 60°\}$, and $\gamma = 0°$ (Fig. 5e, f). Additionally, the geometric relationship between the forelimb of the robot and its torso scapula, at $\theta = 15°$ (Fig. 4a), must be considered. This relationship is observed not only in actual mole structures but also facilitates more effective burrowing by the robot at that angle [31]. Results indicate that granular resistance on each forepaw decreases as β increases, with F_z and F_x approaching similar values at $\beta = 60°$. The CMF and EF forepaws exhibit less granular resistance than the other three forepaws, with the EF showing significantly lower resistance than the CMF.

In Fig. 5, the predicted F_z for the EF is the lowest across different β angles, indicating its suitability for designing burrowing modules. However, the EF also generates relatively low horizontal thrust, which necessitates further optimization. Consequently, a series of dimensional constraints

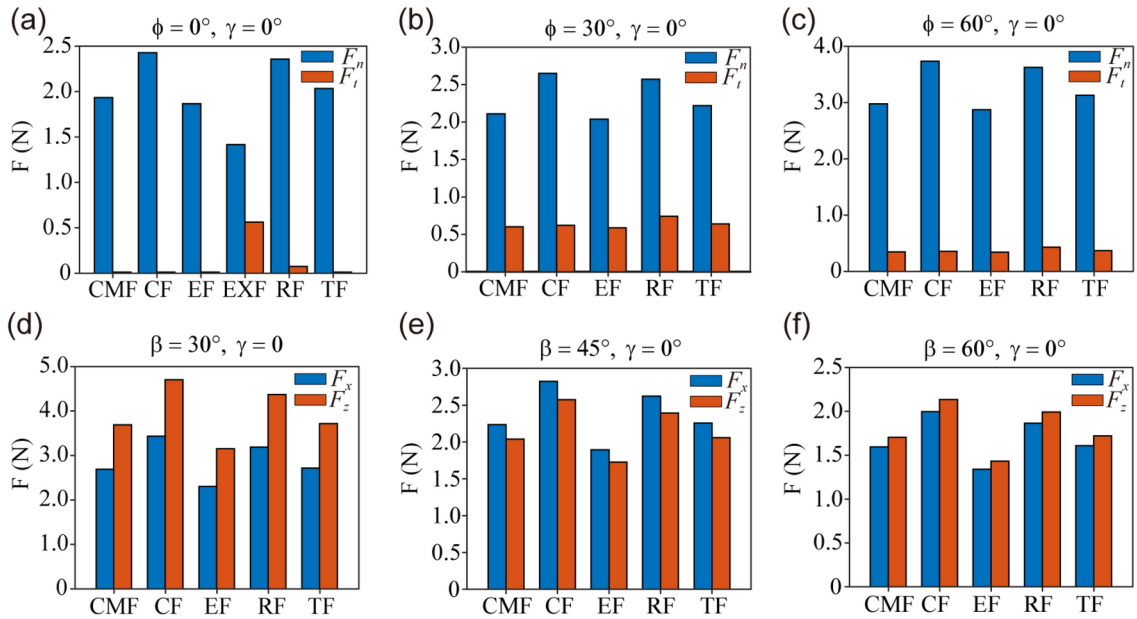

Fig. 5 Granular resistance acting on the forepaw of the robot. **a** The normal force (F_n) and tangential force (F_t) generated during horizontal motion at three angles of $\phi \in \{0°, 30°, 60°\}$ relative to the direction of velocity; **b** the lift force F_z and horizontal force F_x acting on the forepaw due to the granules when $\beta \in \{30°, 45°, 60°\}$ and $\gamma = 0°$

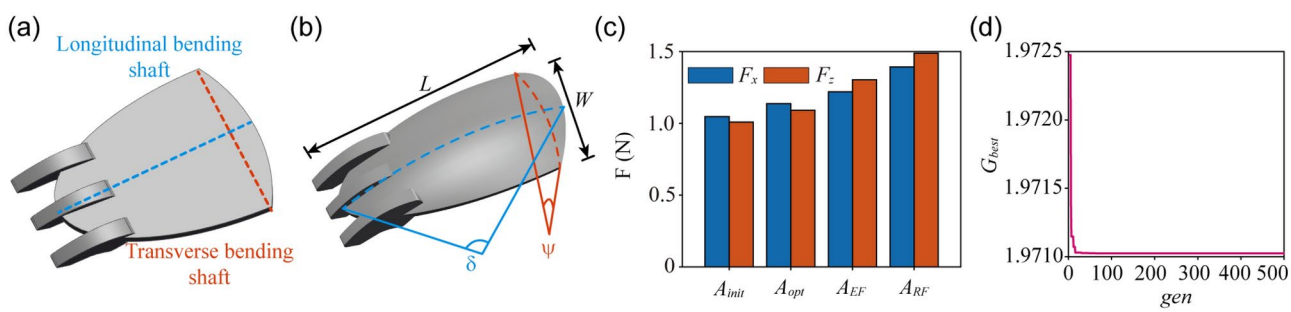

Fig. 6 Geometric definition of the EF. **a** The blue and orange dashed lines indicate the longitudinal and transverse bending axes, respectively; **b** Angles δ and ψ represent the degrees of longitudinal and transverse bending, respectively; **c** Comparison of granular resistance before and after optimization; **d** Iterative process of G_{best}

were implemented, leading to additional parameter optimization to determine the optimal configuration for the robotic forelimb.

3.3 Elliptical Forepaw Optimization

As detailed in Sect. 3A, we incorporated parameters for longitudinal and transverse curvature into the EF to identify optimal configurations. In Fig. 6, the length L is defined as the long axis of the ellipse, while the width W is twice the short axis. The blue and orange dashed lines indicate the longitudinal and transverse bending axes, respectively. The degree of longitudinal and transverse bending are denoted by angle δ and angle ψ. Each parameter is outlined in Table 2.

Building on the previous discussion, the optimization targets for the EF include its length, width, longitudinal curvature, and transverse curvature, represented as $A = [L, W, \delta, \psi]$. We employ the established RFT to calculate granular resistance across different parameters, aiming to identify those configurations that yield the maximum F_x while adhering to specified constraints. This formulation presents a nonlinear multivariate optimization problem with the object of achieving G_{best}. In the absence of specific requirements, we set the initial values to $A_{init} = [45, 35, 70°, 30°]$, indicating that the optimization starts from the median of the specified ranges. The optimization process is outlined in the following algorithm:

Input: $\alpha_{x,z}(\beta,\gamma)$, $|z|$ {RFT and depth parameter};
Output: $\mathbf{A} = [L, W, \delta, \psi]$;
1: Set the optimization target $G_{best}\{F_x - F_z\}$ and optimization algorithm ζ;
2: Set parameter constraints;
3: Initialize $\mathbf{A} = [45, 35, 70°, 30°]$ and population velocity V {Initialization parameters};
4: Set the population size N and the number of iterations g;
5: **Repeat**
6: Calculate z, v, ds, df_z and df_x for each individual P_i;
7: Compute the total Force: $F_i = \sum df_i$ $(i = x,\ z)$ for each P_i and update P_{ibest};
8: Update A and G_{best} by optimization algorithm;
9: **Until** ζ converges

Algorithm Elliptical Forepaw Optimization Framework

Firstly, we set the RFT parameters $\alpha_{x,z}(\beta,\gamma) = \alpha_{x,z}(60°, 0°)$ and the depth of the EF center of mass to $|z| = 3$ cm. The parameter optimization algorithm for the forepaw employs the PSO algorithm ζ, with the optimization target defined as $G_{best} = F_x - F_z$. Based on the previously established parameter constraints, we initialize the population speed range $V \in [-1.5, 1.5]$ and set the number of iterations to $g = 500$, with a total population of $N = 100$ individuals. Granular resistance is calculated using Eq. 1 to determine the total F_z and F_x for each individual P_i. The individual fitness value P_{ibest} is updated based on these calculations, followed by iterative updates of the parameters for each P_i and the overall population fitness G_{best}. This process is repeated until the optimization algorithm ζ converges, yielding in the optimized parameters $A = [L, W, \delta, \psi]$.

Next, we provide an optimization example for the EF, aiming to achieve lower F_z and higher F_x. The optimization algorithm is implemented in a MATLAB environment. Using the initial parameters $A_{init} = [45, 35, 70°, 30°]$ as a reference for comparison, the results is illustrated in Fig. 6c. The iterative process for G_{best} is depicted in Fig. 6d. Initially, the calculated F_x for the forepaw exceeds F_z compared with the previous results in Fig. 5b, with $F_x = 1.0466N$ and $F_z = 1.0088N$ after applying bending constraints. The optimization algorithm yields $A_{opt} = [45.652, 38.022, 64.7771°, 32.6335°]$, resulting in $F_x = 1.1373N$, an increase of 8.67%, and $F_z = 1.0917N$, an increase of 8.22%. This optimized forepaw achieves a balance between increased F_x and reduced F_z, exemplifying a multi-objective optimization scenario. However, the primary focus remains on enhancing F_x to meet the propulsion requirements of the robotic system while maintaining a low F_z.

Fig. 7 Forepaws used in this study, marked with their respective names

3.4 Experimental Verification

To evaluate the performance of the optimized forepaw in GM, we utilized the mole-inspired forelimb-burrowing robot shown in Fig. 4a. This robot measures 21.2 cm in length, 18.5 cm in width, and 5.5 cm in height. It is equipped with a head, a forelimb burrowing mechanism, a rigid spine, and a hindlimb mechanism. Except for the drive motor and rigid optical axis, all components are constructed from 3D-printed ABS resin. The forelimb is designed with a four-bar mechanism and powered by an N20 DC geared motor, which operates at 14 RPM with a nominal torque of 0.32 N m. The trajectory of the forelimb end mimics the burrowing motion of a mole (Fig. 4a). For testing purposes, We 3D printed three types of forepaws: Rectangular Forepaw (RFO), Elliptical Forepaw (EFI), and Optimized Elliptical Forepaw (EFO) (Fig. 7).

The size of the robot forepaw was set to match the optimization result A_{opt}. Unlike the other forepaws, which lacked curvature settings, the EFO incorporated these parameters. Each forepaw was tested under identical burrowing conditions. We conducted two experiments to evaluate the performance of the forepaws. In the robot force-loading test, the head mechanism of the robot was removed, and the forelimb was connected to a 3D force sensor (LA3D40), which was fixed at the end of a six-axis robotic arm (UR3). The robotic forelimb was then allowed to move at a consistent depth ($|z| = 3$ cm) beneath the granular bed, with the robotic arm remained stationary. The PC recorded the force data from the robot via the sensor at a frequency of 100 Hz (Fig. 8a). In the robot burrowing test, we assembled a complete prototype, excluding the head mechanism, and placed it in a granular bed measuring 900 mm × 400 mm × 130 mm. Three types of forepaw burrowing experiments were conducted at the same depth, with data on movement distance

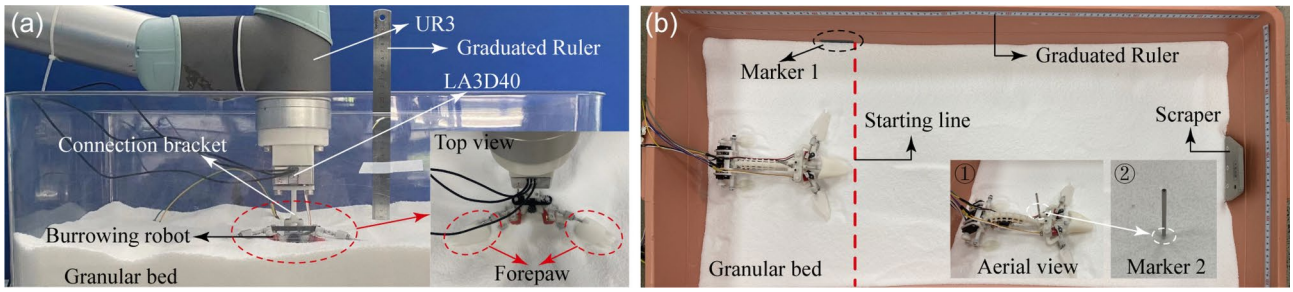

Fig. 8 Two robotic experiments conducted in this study. **a** Robot force-loading test in GM; **b** Robot burrowing test in GM

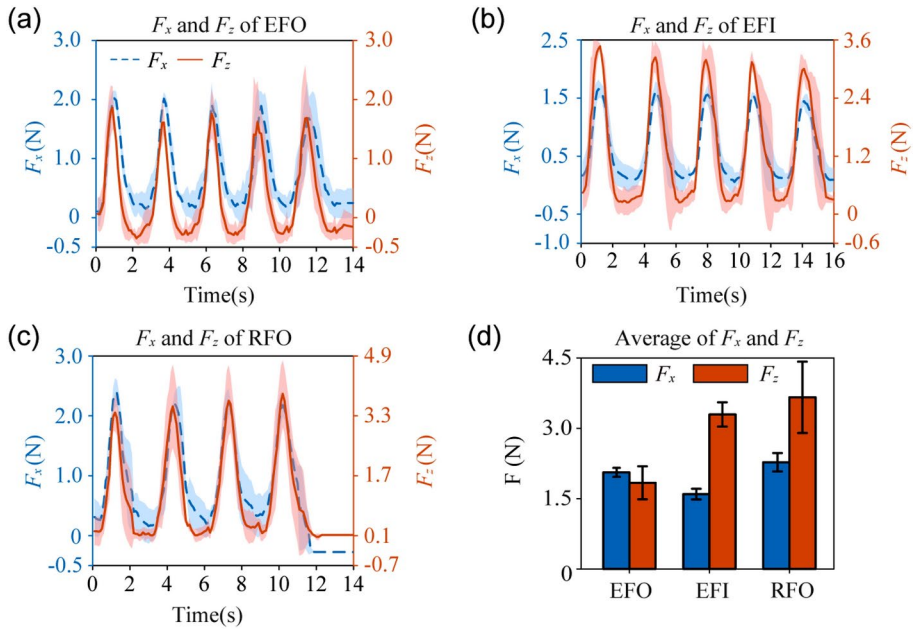

Fig. 9 Results from the robot force-loading tests in granular media, where solid and dotted lines represent the mean values over five trials, and the shaded area indicates the $+/-$ standard deviation of the results. **a**–**c** Forces F_x and F_z for the robot equipped with different forepaws; **d** Mean and standard deviation of the maximal values of F_x and F_z for the robot with different forepaw designs

and attitude recorded (Fig. 8b). The robot was initially positioned at the starting point (Marker 1) and then buried to a fixed depth of $|z| = 4$ cm (Marker 2) to ensure complete coverage by the granules. Additionally, the head mechanism was removed to eliminate any influence of head angle on the overall granular force experienced by the robot. To ensure accuracy, the robot was equipped solely with forelimb and hindlimb mechanisms during the tests.

We conducted force-loading tests on three different forepaws, collecting data once the robot reached a steady burrowing state. Each trial was performed more than five times (Fig. 9a–c), and the granular bed was manually reset with a brush between trials. During the steady burrowing phase, we collected data to calculate the mean and standard deviation (Fig. 9d). The results indicated that the F_z values for both the RFO and EFI were greater than that of the EFO, with increases of 99.30% and 79.44%, respectively. This finding aligns with the RFT calculations, suggesting that the EFO exhibits superior resistance to lift force. In terms of F_x , while the EFO generated 10.49% less thrust than the RFO, it still produced 22.55% more thrust than the EFI. This indicates that the EFO effectively maintains a larger F_x for robotic movement while simultaneously reducing F_z. Notably, the actual force-loading data from the three forepaws were approximately double the theoretical values (Fig. 6c), with all forepaws experiencing greater F_z in experiments compared with theoretical predictions. This discrepancy arises because, during burrowing, the granules displaced by the forepaw accumulate in the middle of the robot torso, resulting in an increase in F_z by the sensor. However, this accumulation does not affect the comparative analysis of the burrowing performance.

We conducted burrowing experiments in GM using three different forepaws, measuring the horizontal displacement of the robot and the time taken from the start of movement until it emerged from the granular surface. Each forepaw was tested across ten trials, with the granular bed evenly scraped between trials. As shown in Fig. 10a, c, the robot equipped with the EFO achieved the largest average horizontal displacement of 179.528 mm, while the RFO lifted less than 50 mm, indicating that the EFO effectively reduces granular lift forces compared with the other forepaws. To

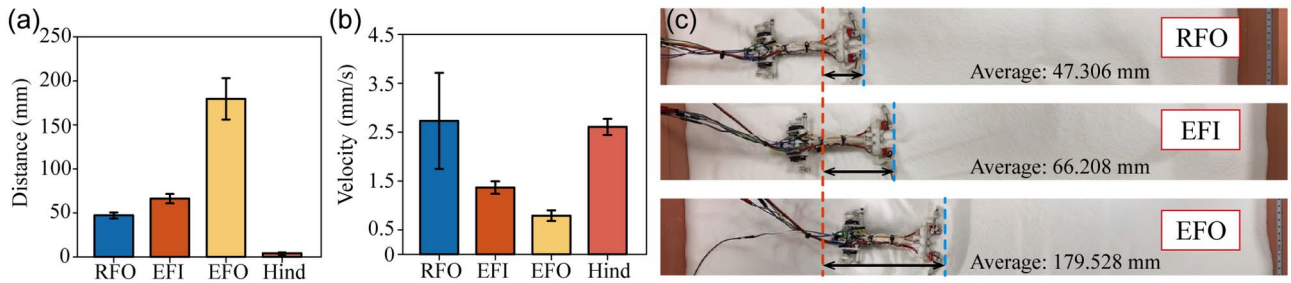

Fig. 10 Results from the robot burrowing tests in granular media. Experimentally measured data represent means over 10 experimental trials. **a** Horizontal displacement of the robot as it burrows to the granular surface; **b** rising speed of the robot; **c** burrowing process of the robot with different forepaws

further assess the performance of the optimized forepaw, we calculated the robotic rise speed under each forepaw condition (Fig. 10b). All robots were buried at the same depth, and movement times were recorded for speed calculations. The robot with the EFO exhibited the lowest rising velocity at 0.9355 mm/s, compared with 3.2311 mm/s for the RFO and 1.6182 mm/s for the EFI. Additionally, we conducted experiments where the robots moved only using their hindlimbs, comparing these results with data from all three forepaw types (Fig. 10a, b). When the robot operated using the hindlimb mechanism, the measured horizontal displacements remained below 5 mm, indicating minimal impact on overall movement. Moreover, the rising speed of the robot was notably high during hindlimb movement. Observations revealed that the hindlimb mechanism often cleared the granular surface first, followed by the rest of the robot. This phenomenon occurs because the hindlimb mechanism, with its single-degree-of-freedom motion, generates a steep trajectory during scraping and burrowing, leading to significant lift forces. The longer torso of the robot creates a leverage effect, allowing the hindlimb to exit the granular surface first, while exerting a limited influence on the forelimb motion.

In conclusion, the optimized forepaw (EFO) demonstrates superior resistance to granular forces, effectively reducing lift force and enhancing the movement capability of the robot. The results confirm that employing the RFT as a design framework for parametric optimization of the burrowing module allows for precise quantification of its performance, thereby improving the mechanical efficacy of the burrowing robot.

4 Conclusion and Future Work

This work demonstrates the capability to analyze the force-loading performance of a burrowing module using the horizontal thrust F_x and vertical lift F_z derived from the RFT model as quantitative performance indices, leading to a design solution. A design framework for mole-inspired burrowing modules is provided, with the robotic forepaw optimized for minimal granular lift and maximal propulsion. Experimental results indicate that the optimized forepaw reduces granular lift force by 99.30% and 79.44% compared with rectangular and elliptical forepaws, respectively. In burrowing tests, the robot equipped with the optimized forepaw achieved the largest displacement of 179.528 mm and the slowest rise speed of 0.9355 mm/s, confirming the effectiveness of the forepaw in enhancing burrowing performance.

The burrowing module design method for mole-inspired robots presented in this work provides valuable guidance for actuator parameter design in similar robotic applications, addressing the challenges associated with slow and inefficient burrowing. Additionally, this design approach can be employed to optimize the structural parameters of key soil-contacting mechanisms, enabling a globally optimal design of the robotic burrowing system. This comprehensive parameter optimization is expected to significantly enhance burrowing efficiency, and we anticipate, in the future, burrowing robots will achieve autonomous submerged movement in GM.

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Data Availability The datasets generated and/or analyzed during the current study are available from the corresponding author on reasonable request.

Declarations

Conflict of interest There are no known competing financial interests or personal relationships that could affect the work reported in this article.

References

1. Jia, H., Wang, Q., Huang, D., Zhu, L., Li, M., & Zhao, J. (2019). Design of bionic mole forelimb intelligent row cleaners. *International Journal of Agricultural and Biological Engineering,12*(3), 27–35.
2. Zhang, T., Xu, K., Yao, Z., Ding, X., Zhao, Z., Huo, X., Pang, Y., Lai, X., Zhang, W., Liu, S., & Deng, J. (2019). The progress of extraterrestrial regolith-sampling robots. *Nature Astronomy,3*, 487–497.
3. Zhang, T., Pang, Y., Zeng, T., Wang, G., Yin, S., Xu, K., Mo, G., Zhang, X., Wang, L., Yang, S., Zhao, Z., Qin, J., Gong, J., Zhao, Z., Tong, X., Yin, Z., Wang, H., Zhao, F., Zheng, Y.,… Ding, X. (2023). Robotic drilling for the Chinese Chang'E 5 lunar sample-return mission. *The International Journal of Robotics Research,42*(8), 586–613.
4. Wei, H., Zhang, Y., Zhang, T., Guan, Y., Xu, K., Ding, X., & Pang, Y. (2021). Review on bioinspired planetary regolith-burrowing robots. *Space Science Reviews,217*(87), 1–39.
5. Zhang, T., Wang, B., Wei, H., Zhang, Y., Chao, C., Xu, K., Ding, X., Hou, X., & Zhao, Z. (2021). Review on planetary regolith-sampling technology. *Progress in Aerospace Sciences,127*, 100760.
6. Kobayashi, T., Tshukagoshi, H., Honda, S., & Kitagawa, A. (2011). Burrowing rescue robot referring to a mole's shoveling motion. In *Proceedings of the 8th JFPS International Symposium on Fluid Power* (pp. 644–649), Ookayama Tokyo, Japan.
7. Tirtawardhana, C., Yu, B., Myeong, W., & Myung, H. (2020). Development of integrated digging robot using drill bit-limbs hybrid mechanism that mimics burrowing animals behavior. In *Proceedings of the 6th International Conference on Robot Intelligence Technology and Applications (RITA)* (pp. 323–332), Putrajaya, Malaysia.
8. Lee, J., Tirtawardhana, C., & Myung, H. (2020). Development and analysis of digging and soil removing mechanisms for mole-bot: Bio-inspired mole-like drilling robot. In *2020 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)* (pp. 7792–7799), Las Vegas, NV, USA.
9. Maladen, R. D., Ding, Y., Li, C., & Goldman, D. I. (2009). Undulatory swimming in sand: Subsurface locomotion of the sandfish lizard. *Science,325*(5938), 314–318.
10. Li, C., Zhang, T., & Goldman, D. I. (2012). *A Resistive Force Model for Legged Locomotion on Granular Media. Adaptive Mobile Robotics* (pp. 433–440). World Scientific.
11. Zhang, T., & Goldman, D. I. (2014). The effectiveness of resistive force theory in granular locomotion. *Physics of Fluids,26*(10), 101308.
12. Thoesen, A., McBryan, T., Green, M., Mick, D., Martia, J., & Marvi, H. (2020). Revisiting scaling laws for robotic mobility in granular media. *IEEE Robotics and Automation Letters,5*(2), 1319–1325.
13. Vogel, S. (2020). *Life in Moving Fluids: The Physical Biology of Flow* (2nd ed., pp. 300–400). Princeton University Press.
14. Maladen, R. D., Ding, Y., Umbanhowar, P. B., Kamor, A., & Goldman, D. I. (2011). Mechanical models of sandfish locomotion reveal principles of high performance subsurface sand-swimming. *Journal of The Royal Society Interface,8*(62), 1332–1345.
15. Li, C., Zhang, T., Zhang, Y., & Goldman, D. I. (2013). A terradynamics of legged locomotion on granular media. *Science,339*(6126), 1408–1412.
16. Agarwal, S., Karsai, A., Goldman, D. I., & Kamrin, K. (2021). Surprising simplicity in the modeling of dynamic granular intrusion. *Science Advances,7*(17), eabe0631.
17. Huh, T. M., Cao, C., Aderibigbe, J., Moon, D., & Stuart, H. S. (2023). Walk-burrow-tug: Legged anchoring analysis using rft-based granular limit surfaces. *IEEE Robotics and Automation Letters,8*(6), 3796–3803.
18. Sandefur, M. (2008). *BiomechanicalModeling ofForelimb Adduction in theEasternMole*, *Scalopusaquaticus*. PhD thesis. Western Kentucky University (pp. 2–25), Bowling Green, USA.
19. Lin, Y.F., Konow, N., & Dumont, E. R. (2019). How moles destroy your lawn: The forelimb kinematics of eastern moles in loose and compact substrates. *Journal of Experimental Biology,222*(4), jeb182436.
20. Liu, C.Y. (2008). *Scaptochirus Moschatus* Cutting Mechanism of Claws of the Mole-rat (*Scaptochirus moschatus*). PhD thesis, Jilin University (pp. 89–101), Changchun, China.
21. Ren, L. Q. (2009). Progress in the bionic study on anti-adhesion and resistance reduction of terrain machines. *Science in China Series E: Technological Sciences,52*, 273–284.
22. Koch, K., Bhushan, B., & Barthlott, W. (2009). Multifunctional surface structures of plants: An inspiration for biomimetics. *Progress in Materials Science,54*(2), 137–178.
23. Tong, J., Sun, J., Chen, D., & Zhang, S. (2005). Geometrical features and wettability of dung beetles and potential biomimetic engineering applications in tillage implements. *Soil and Tillage Research,80*(1–2), 1–12.
24. Tong, J., Moayad, B. Z., Ren, L. Q., & Chen, B. C. (2004). Biomimetics of soft terrain machines: A review. *Korean Society of Agricultural Machinery,9*(2), 19–40.
25. Ji, W., Chen, D., Jia, H., & Tong, J. (2010). Experimental investigation into soil-cutting performance of the claws of mole rat (*Scaptochirus moschatus*). *Journal of Bionic Engineering,7*, S166–S171.
26. Zhang, Z., Jia, H., & Sun, J. (2016). Review of application of biomimetics for designing soil-engaging tillage implements in Northeast China. *International Journal of Agricultural and Biological Engineering,9*(4), 12–21.
27. Ji, W., Tong, J., Jia, H., Chen, D., & Liu, C. (2010). Quantitative characteristic features of geometric structures of claws of mole rat. *Transactions of the CSAM,41*(4), 193–198.
28. Ji, W., Jia, H., & Tong, J. (2012). Experiment on working performance of bionic blade for soil-rototilling and stubble-breaking. *Transactions of the Chinese Society of Agricultural Engineering,28*(12), 24–30.
29. Scott, R.G., & Richardson, R.C. (2005). Realities of biologically inspired design with a subterranean digging robot example. In *Proceedings of the 6th IASTED international conference on robotics and applications* (pp. 226–231), Cambridge, MA, USA.
30. Xu, L., Zhang, S., Jiang, N., & Xu, R. (2015). A hybrid force model to estimate the dynamics of curved legs in granular material. *Journal of Terramechanics,59*, 59–70.
31. Zhang, T., Wei, H., Zheng, H., Liang, Z., Yang, H., Zhang, Y., Zhu, H., Guan, Y., Ding, X., Wang, K., & Xu, K. (2024). Mole-inspired robot burrowing with forelimbs for planetary soil exploration. *Advanced Intelligent Systems,6*(6), 2300392.

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